

IMO(34th) – 1993

First Day

1. Let $n > 1$ be an integer and let $f(x) = x^n + 5x^{n-1} + 3$. Prove that there do not exist polynomials $g(x), h(x)$ each having integer coefficients and degree at least one, such that $f(x) = g(x)h(x)$.
2. A, B, C, D are four points in the plane, with C, D on the same side of the line AB , such that $AC \cdot BD = AD \cdot BC$ and $\angle ADB = 90^\circ + \angle ACB$. Find the ratio $\frac{AB \cdot CD}{AC \cdot BD}$ and prove that circles ACD, BCD are orthogonal. (Intersecting circles are said to be orthogonal if at either common point their tangents are perpendicular.)
3. On an infinite chessboard, a solitaire game is played as follows: At the start, we have n^2 pieces occupying n^2 squares that form a square of side n . The only allowed move is a jump horizontally or vertically over an occupied square to an unoccupied one, and the piece that has been jumped over is removed. For what positive integer n can the game end with only one piece remaining on the board?

Second Day

4. For the three points A, B, C in a plane we define $m(ABC)$ to be the smallest length of the three altitudes of the triangle ABC , where in the case of A, B, C collinear, $m(ABC) = 0$. Let A, B, C be given points in the plane. Prove that for any point X in the plane, $m(ABC) \leq m(ABX) + m(AXC) + m(XBC)$.
5. Let $N = \{1, 2, 3, \dots\}$. Determine whether there exists a strictly increasing function $f : N \rightarrow N$ with the following properties:
 - (a) $f(1) = 2$
 - (b) $f(f(n)) = f(n) + n, \quad (n \in N)$
6. Let n be an integer greater than 1. In a circular arrangement of n lamps $L_0, L_1, \dots, L_{n-1}$, each one of that can be either on or off, we start with the situation where all lamps are on, and then carry out a sequence of steps, $Step_0, Step_1, \dots$. If L_{j-1} (indices are taken modulo n) is on, then $Step_j$ changes the status of L_j (it goes from on to off or from off to on) but does not change the status of any other lamps. If L_{j-1} is off, then $Step_j$ does not change anything at all. Show that:
 - (a) There is a positive integer $M(n)$ such that after $M(n)$ steps all lamps are on again.
 - (b) If n has the form 2^k , then all lamps are on after $n^2 - 1$ steps.
 - (c) If n has the form $2^k + 1$, then all lamps are on after $n^2 - n + 1$ steps.

